

Litt review Cath (NEWEST VERSION MA ZO)

Credit risk assessment plays a central role in the financial sector, as lenders must constantly evaluate whether a borrower is likely to repay a loan or default on their obligations. To support these decisions, financial institutions rely on credit scoring models, which estimate the probability of default based on financial, behavioural, and demographic information. With the growth of data availability and computational power, machine learning has become increasingly important in credit scoring. Therefore, research has progressively shifted beyond predictive accuracy alone toward questions of robustness, interpretability, and fairness using feature attributions such as SHAP and LIME.

For instance, Bückner et al. explore explainability in credit scoring through the TAX4CS framework, which organises around three main dimensions: transparency, auditability, and explainability. In their analysis, the authors particularly focus on feature attribution methods such as SHAP, LIME, and iBreakDown, which identify how individual variables contribute to a credit decision. The paper also discusses global explanation techniques, including permutation feature importance and partial dependence plots, used to analyse overall model behaviour and verify whether learned relationships are economically plausible.

The authors show that traditional approaches such as scorecards and logistic regression remain widely used because they are easier to interpret and audit than many black box machine learning models. They further emphasize that, in credit scoring, predictive performance alone is insufficient; models must also remain transparent, robust, and trustworthy. This perspective is closely related to Influence Functions, which complement feature attribution methods by identifying which training examples influenced the model's behaviour and predictions.

Additionally, Provenzano et al. (2020) also propose a machine learning framework for credit scoring and default prediction that combines heterogeneous data sources, including structured financial variables and unstructured text data.

To improve the reliability of the predictions, the authors apply probability calibration so that the estimated probabilities of default better align with observed default frequencies. They then transform these calibrated probabilities into an internal credit rating system using a genetic algorithm, while enforcing constraints such as monotonicity and cluster quality to ensure consistency and interpretability of the rating structure.

In addition to predictive performance, the authors incorporate explainability techniques by applying SHAP and LIME to provide local explanations of individual credit decisions. This method allows the model's outputs to be interpreted at the level of single applicants, which is particularly important in credit risk management and regulatory contexts.

However, Ratul, et al. (2021) have found that both SHAP and LIME exhibit important limitations in practice. Their evaluation shows a lack of precision, as features identified as important for one class often also appear in the opposite class. The authors find that both SHAP and LIME exhibit important limitations in practice. Their evaluation shows a lack of precision, as features identified as important for one class often also appear in the opposite class. They also observe weak generality, since explanations vary significantly across similar

instances within the same class, and low consistency, with substantial disagreement between SHAP and LIME in terms of feature ranking. Overall, the study questions the reliability of these widely used attribution methods, suggesting that their explanations may not always be stable or representative of underlying class-level patterns.

Overall, the literature on credit scoring has largely relied on feature attribution methods such as SHAP, LIME, and partial dependence plots to interpret model predictions and explain the influence of input variables on credit decisions.

Therefore, data attribution such as influence function has been implemented. The concept of the influence curve has been introduced by Hampel (1974), now known as the influence function, as a foundational tool in robust statistics. It is defined as the derivative of a statistical estimator viewed as a functional, and measures the effect of an infinitesimal contamination at a given point on the value of the estimator. The author shows that this framework can be used to study local robustness, derive asymptotic variances, quantify sensitivity to outliers through gross-error sensitivity, and support the construction of robust estimators such as M-estimators and trimmed means.

This paper is the origin of influence functions as used in modern machine learning such as Koh and Liang (2017) for explaining black-box predictions, Koh et al. (2019) for group effects, Basu et al. (2021) for evaluating their fragility in deep learning extend back directly to Hampel's influence curve. The influence function used today in ML is the finite-sample version of Hampel's theoretical curve, computed at the empirical distribution.

By extending, Sharchilev et al. (2018) tackle the problem of identifying which training samples most influence the predictions of Gradient Boosted Decision Trees (GBDTs). Since classical influence-function methods (Koh & Liang, 2017) rely on smooth, differentiable models, they cannot be applied directly to tree ensembles, whose structure is discontinuous. The authors provide the first adaptation of influence-function ideas to GBDTs by exploiting the fact that, although tree splits are non-differentiable, leaf values change smoothly when training sample weights are perturbed.

Their contribution is twofold. First, they introduce LeafRefit and FastLeafRefit, which approximate leave-one-out retraining by keeping the tree structure fixed and recomputing leaf values efficiently. Second, they derive LeafInfluence and FastLeafInfluence, the first methods to compute gradients of GBDT predictions with respect to training-sample weights, enabling infinitesimal influence estimation analogous to classical influence functions. These techniques allow the model to estimate how much a training point affects a test prediction without retraining the entire ensemble, making influence analysis computationally feasible for large datasets.

To conclude, Sharchilev et al. (2018) provide the first influence-function framework for tree-based models, using the leaf structure of GBDTs to approximate leave-one-out effects and identify influential training samples efficiently, a direct precursor to later methods such as BoostIn.

(référence du BOOST IN de developed by Brophy et al. in their paper)

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